Activity 2

Ankita Bhattacharyya and Cayla Stevenson

2026-03-08

Introduction and Background

The goal of this analysis to be able to build a predictive model for insurance prices using peoples demographic and characteristics in the "insurance data" dataset. The characteristics in the dataset include variables like age, sex. BMI, number of kids, if they smoke, and more. We used linear regression and other techniques learned in class to find out which variables are most important for insurance price prediction. We also used two different models to complete this task and compare results to determine which one performed the best. The first method we used was subset selection algorithms such as forward selection with AIC and backward selection with BIC. The second method we used was LASSO regression. The goal of this report is to view results and determine which model gives the most reliable predictions of insurance charges.

Data Exploration

The dataset includes information on insurance policy holders and predictors/ characteristics that impact insurance price. The predictor variables include the individuals demographic and characteristics related to health such as age, bmi, number of children, smoking, region, month, and any prior claims the individual may have.

The chosen variables for the dataset were used to locate any patterns that may be associated with the data. We noticed in the dataset that there is a relationship between people that smoke and a higher insurance charge. We also noticed that the BMI variable had some correlation with higher insurance charges meaning that the higher the BMI, the higher the insurance cost possibly because a higher BMI could lead to more health issues or risks. Overall, the main things we noticed is that there are some predictors in the dataset like smoking and BMI that may impact the price of insurance the most.

Modeling Methods

Method One: Subset Selection Algorithms

To find variables for the model, two subset selection algorithms were used; Forward Selection with the performance indicator AIC and Backward Selection with the performance indicator BIC.

Each gave difference variables to be included in the model. The backwards selection included fewer predictors than the forward selection did. To select between the two models, we did some further analysis on certain predictors in each model and the checked for violations of multi-linear assumptions.

The two base models that are being analyzed are:

Forward AIC (Model A):

$\widehat{charges} = -11437.82 + 23045.34smokeryes + 259.86age + 338.86bmi - 617.69regionnorthwest - 1950.74regionsoutheast - 1512.27regionsouthwest + 528.11children + 346.21priorclaims - 106.75month$

Backward BIC (Model B):

$$\widehat{charges} = -11627.90 + 257.65age + 310.57bmi + 502.99children + 23016.60smokeryes$$

Significance of Predictors

When conducting a hypothesis test of significance for each predictor in Model A, `region`'s t statistic and p-value when `region` is northwest and all other predictors are constant are -1.013 and 0.31123, respectively. Under a 5% significance level, there isn't enough evidence to say region is a significant enough predictor. The predictor is removed from the model, and the resulting model as: $\widehat{charges} = -11612.23 + 22996.69smokeryes + 261.46age + 305.57bmi + 538.87children + 359.47priorclaims - 95.95months$

Continuing to remove insignificant predictors, when conducting another hypothesis test for significance for each predictor in Model A, `month`'s t-statistic and p-value are -1.547 and 0.12216 respectively. Under a 5% significance level, there isn't enough evidence to month is a significant enough predictor. Once removed the model becomes:

$\widehat{charges} = -12154.03 + 22984.80smokeryes + 260.38age + 304.55bmi + 530.38children + 355.36priorclaims$

For the resulting Model A; 72.05 % of the model's variation in charges is explained by predictors of smoker, age, bmi, children and prior claims.

For Model B, when we did hypothesis tests for significance, all predictors in the model were significant at the 5% level. So no predictors needed to removed.

Testing Multi-linearity Assumptions

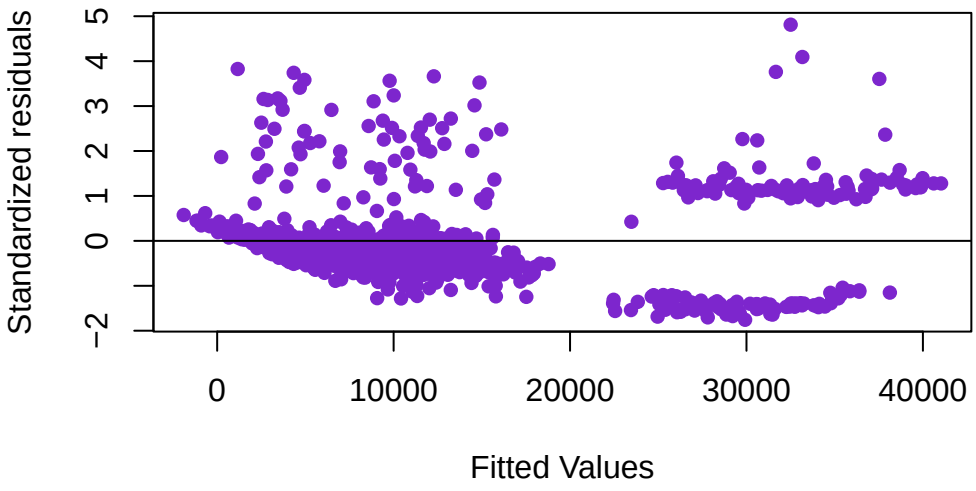
Checking for departures from linearity in both models.

For both models, since our data was taken by random sample, so we can assume that independence assumption has not been violated.

Model A Assumptions

For Model A, to check linearity, constant error variance and zero mean, we created a Standardized residuals plot.

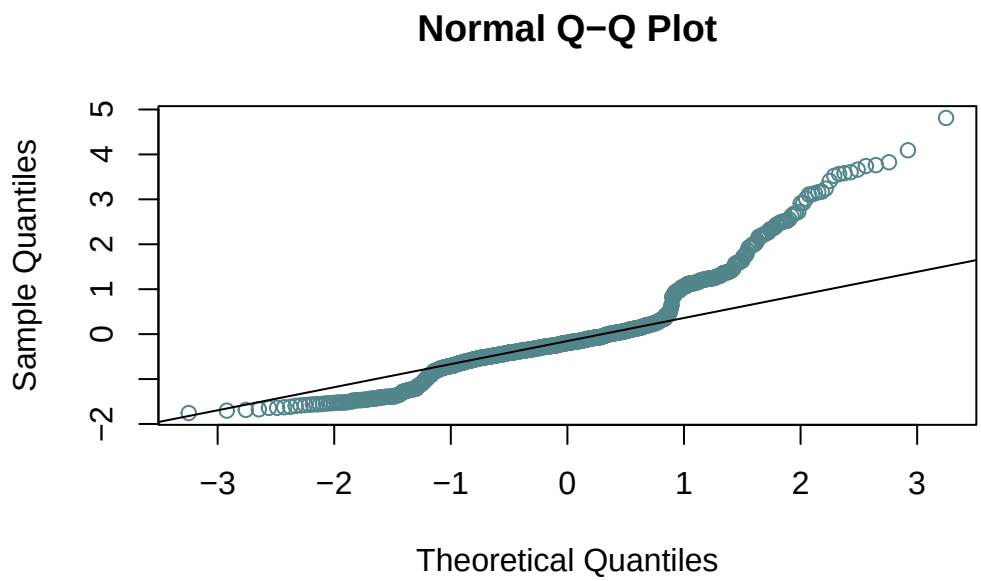
Figure 1: Standardized Residual vs Fitted Values plot for Model A



The Residual plot shows no apparent curvature and no fanning, so it is safe to assume there are no violations to linearity and there is constant error variance. The majority of points are clustered around the zero mean, with two clusters clearly above and below the zero mean line. So there is a possible violation of zero mean distribution.

Next, to check if our distribution is normal, a quantile-quantile plot is created.

Figure 2: Quantile-Quantile plot for Model A

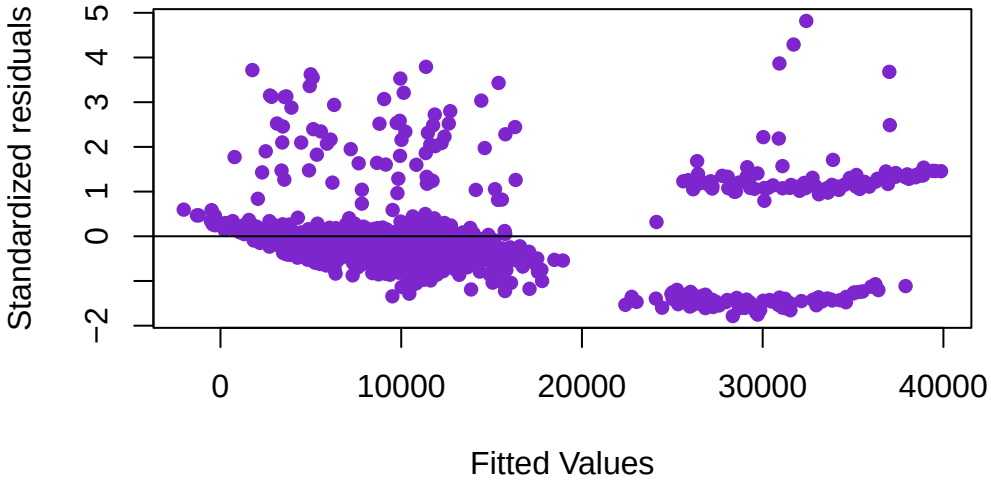


The distribution seems to be heavily right skewed, since the actual values departure from the theoreticals quantities on the right of the graph.

Model B Assumptions

To check linearity, constant error variance and zero mean, we created a Standardized residuals plot.

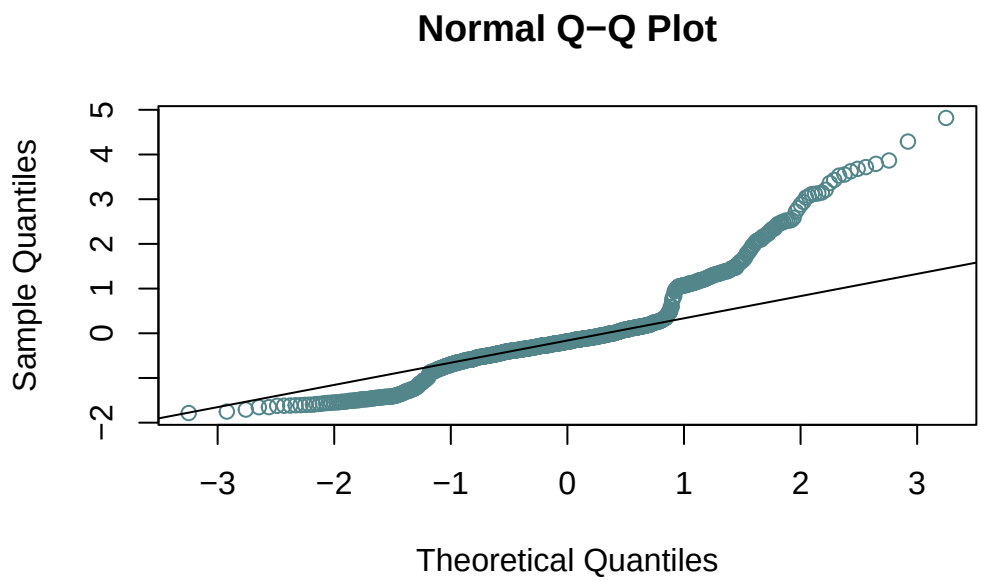
Figure 3: Standardized Residual vs Fitted Values plot for Model B



The residual plot shows no apparent curvature and no fanning, so it is safe to assume there are no violations to linearity and there is constant error variance. The majority of points are clustered around the zero mean, with two clusters clearly above and below the zero mean line. So there is a possible violation of zero mean distribution. Almost exactly like Model A.

Next, to check if our distribution is normal, a quantile-quantile plot is created.

Figure 4: Quantile-Quantile plot for Model A



The distribution seems to be heavily right skewed, since the actual values departure from the theoreticals quantities on the right of the graph. ### Comparing Performance indicators

Method Two: Lasso Regression

The second modeling approach used LASSO regression. LASSO performs variable selection and regularization by shrinking less important regression coefficients toward zero. The predictors with numbers that shrink to zero are then removed from the model. The model was used to look at the variables age, bmi, number of children, smoking, region, month, and any prior claims. The variables that had shrunk to zero were removed from the model and then the model was evaluated and the RMSE was calculated. I used the glmnet package in R to do cross validation.

The variables retained by the LASSO model included: age, BMI, number of children, smoking, region, month, and prior claims. We removed the predictors/ variables that went to zero and were removed.

Results

Initial Results

Initial models used the algorithms with forward selection with AIC and backward selection with BIC. The forward selection produced a model that had multiple predictors of smoking, age, BMI, region, etc. The backward selection resulted in a more simple model that had only age, BMI, number of children, and their smoking status. Both of the models show that smoking had the most

impact on the insurance charges. The model proved that people that smoke were predicted to have a higher insurance cost vs non smokers.

Interpretations

Model Evaluation

The LASSO model was used with the validation dataset and RMSE was calculated.The RMSE for LASSO was 5225.281 which means that the predicted insurance charges are different than the actual charges. The RMSE gives a measure of the prediction and while some of the predictions may be closer than the actual number, this shows the overall accuracy of the model with this data.

Refinements

Based on the model selection and evaluation, both models produced reasonable models for predicting insurance charges. The backward BIC model resulted in a simpler and more interpretable model with fewer predictors, while LASSO kept the additional predictors.

Model Chosen and Why

Final Reccomendations

Author Contributions and Responsibilities

Ankita Bhattacharyya

Cayla Stevenson

Code Appendix

```
# Creating Models from Subsetting Algorithms-------------------------------
library(tidyverse)
library(Hmisc)
library(modelr)
library(caret)

insurance_data <- read.csv("insurance_data_S26.csv")

insurance_data <- insurance_data %>%
  drop_na()

#FORWARD SELECTION (AIC)----------------------
```

```
null_insurance <- lm(
  charges ~ 1,
  data = insurance_data
)

full_insurance <-lm(
  charges ~ .,
  data = insurance_data
)

stats::step(
  object = null_insurance,
  scope = list(lower = null_insurance, upper = full_insurance),
  data = insurance_data,
  direction = 'forward'
)

FORWARD_AIC <- lm(charges~smoker+age+bmi+region+children+priorclaims+month, data = insurance_da

#Testing Significance predictors------------------------------
#Using terminal output
FORWARD_AIC <- lm(charges~smoker+age+bmi+children+priorclaims+month, data = insurance_data)
FORWARD_AIC <- lm(charges~smoker+age+bmi+children+priorclaims, data = insurance_data)

#plot(FORWARD_AIC) <- looking at the assumptions of linearity

#BACKWARD BIC MODEL ----------------------------------------
stats::step(
  object = full_insurance,
  data = insurance_data,
  direction = "backward",
  k = log(nrow(insurance_data)),
  trace = 1
)

BACKWARDS_BIC <- lm(charges~age+bmi+children+smoker, data = insurance_data)


#CROSS VALIDATION FORWARD AIC ------------------------------

# Cross Validation (80-20 split) -

set.seed(550)
train_index <- sample(
  x = 1:nrow(insurance_data),
  size = floor(0.8*nrow(insurance_data))
)
```

```
insuranceTrain <- insurance_data[train_index, ]
insuranceValid <- insurance_data[-train_index, ]

insuranceTrainModel <- lm(
  formula = charges ~ smoker+age+bmi+children+priorclaims,
  data = insuranceTrain
)

insuranceValid <- insuranceValid %>%
  mutate(
    model1_predict = predict(
      object = insuranceTrainModel,
      newdata = insuranceValid
    )
  )

#CROSS VALIDATION BACKWARD BIC -----------------------------

set.seed(550)
train_index <- sample(
  x = 1:nrow(insurance_data),
  size = floor(0.8*nrow(insurance_data))
)

insuranceTrain <- insurance_data[train_index, ]
insuranceValid <- insurance_data[-train_index, ]

insuranceTrainModel <- lm(
  formula = charges ~ smoker+age+bmi+children+priorclaims,
  data = insuranceTrain
)

insuranceValid <- insuranceValid %>%
  mutate(
    model1_predict = predict(
      object = insuranceTrainModel,
      newdata = insuranceValid
    )
  )
# MODEL A PLOTS --------------------------------------
library(tidyverse)
insurance_data <- read.csv("insurance_data_S26.csv")
insurance_data <- insurance_data %>%
  drop_na()

FORWARD_AIC <- lm(charges~smoker+age+bmi+children+priorclaims, data = insurance_data)
```

```
AResid <- rstandard(FORWARD_AIC)
AFitted <-FORWARD_AIC$fitted.values

plot(AFitted,AResid, xlab = "Fitted Values", ylab = "Standardized residuals", pch = 16, col = "
abline(h=0)
library(tidyverse)
insurance_data <- read.csv("insurance_data_S26.csv")

FORWARD_AIC <- lm(charges~smoker+age+bmi+children+priorclaims, data = insurance_data)

qqnorm(AResid, pch = 1, col = "cadetblue4")
qqline(AResid)
# MODEL A PLOTS -------------------------------------
library(tidyverse)
insurance_data <- read.csv("insurance_data_S26.csv")
insurance_data <- insurance_data %>%
  drop_na()

BACKWARDS_BIC <- lm(charges~age+bmi+children+smoker, data = insurance_data)

BResid <- rstandard(BACKWARDS_BIC)
BFitted <-BACKWARDS_BIC$fitted.values

plot(BFitted,BResid, xlab = "Fitted Values", ylab = "Standardized residuals", pch = 16, col = "
abline(h=0)
library(tidyverse)
insurance_data <- read.csv("insurance_data_S26.csv")
insurance_data <- insurance_data %>%
  drop_na()

BACKWARDS_BIC <- lm(charges~age+bmi+children+smoker, data = insurance_data)

qqnorm(BResid, pch = 1, col = "cadetblue4")
qqline(BResid)
#LASSO MODEL ---------------------------

library(glmnet)
library(tidyverse)

insurance_data <- insurance_data %>%
  drop_na()

train_index <- sample(
  x = 1:nrow(insurance_data),
  size = floor(0.8*nrow(insurance_data))
)
```

```
insuranceTrain <- insurance_data[train_index, ]
insuranceValid <- insurance_data[-train_index, ]

x_train <- model.matrix(charges ~ ., insuranceTrain)[,-1]
y_train <- insuranceTrain$charges

x_valid <- model.matrix(charges ~ ., insuranceValid)[,-1]
y_valid <- insuranceValid$charges


# running lasso
set.seed(550)

lasso_cv <- cv.glmnet(
  x_train,
  y_train,
  alpha = 1
)

#plot(lasso_cv)
```

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